

Constraint-safe quantum-guided portfolio optimization

A scalable classical solver with a safe, auditable quantum proposal interface

EXACT SAFETY

17 guardrail families

MEASURED SCALE

300,000 assets

REAL HARDWARE

8–28 qubits

Portfolio rules matter more than circuit energy

The required output

An investable table of asset weights, trades, risk, income, cost, and a rationale a portfolio team can audit.

01 Improve expected utility

Balance risk, growth, income, drawdown, and cost

02 Respect every guardrail

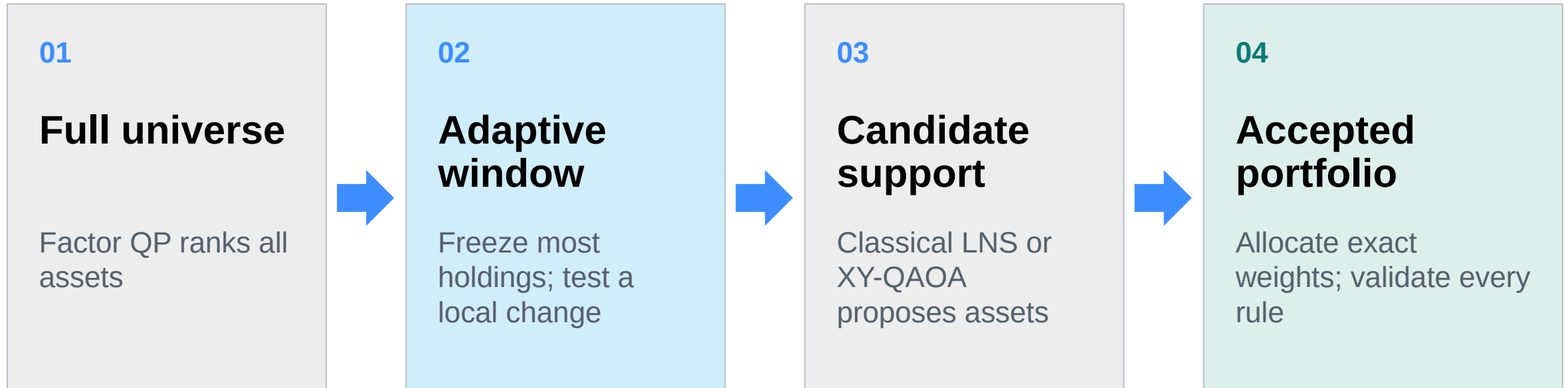
Budget, holdings, groups, turnover, factors, stress, and CVaR

03 Prove what happened

Classical baselines, validation, scalability, explainability, and hardware provenance

Our standard: a poor or noisy proposal is rejected—never repaired into an unsafe recommendation.

One safety boundary keeps every method fair



The quantum computer never assigns capital and never bypasses the independent validator.

The model makes investor trade-offs measurable

min

RISK

 $\lambda_r \cdot \text{variance}$

GROWTH

 $-\lambda_g \cdot \text{return}$

INCOME

 $-\lambda_y \cdot \text{yield}$

COST

 $+\lambda_c \cdot \text{turnover cost}$

Hard guardrails

Capital

100% invested
Long-only
Position min/max

Composition

Exactly K holdings
Eligible + mandatory
Group bands

Risk

Factor bands
Stress floors
95% CVaR limit

Implementation

Turnover cap
Trading cost
Liquidity cap

Lower composite objective is better. It is not a percentage return.

Correctness is triangulated before scale is claimed

 1.41×10^{-9}

objective spread across SciPy,
OSQP, Clarabel, and Gurobi

Exact match

tiny exhaustive enumeration
equals Gurobi at -0.03681145

0.0% gap

100-asset Gurobi certificate at
objective -0.0384147146

Valid initialization

-0.03468764

feasible starting point

LNS / XY-QAOA / penalty QAOA

-0.03804482

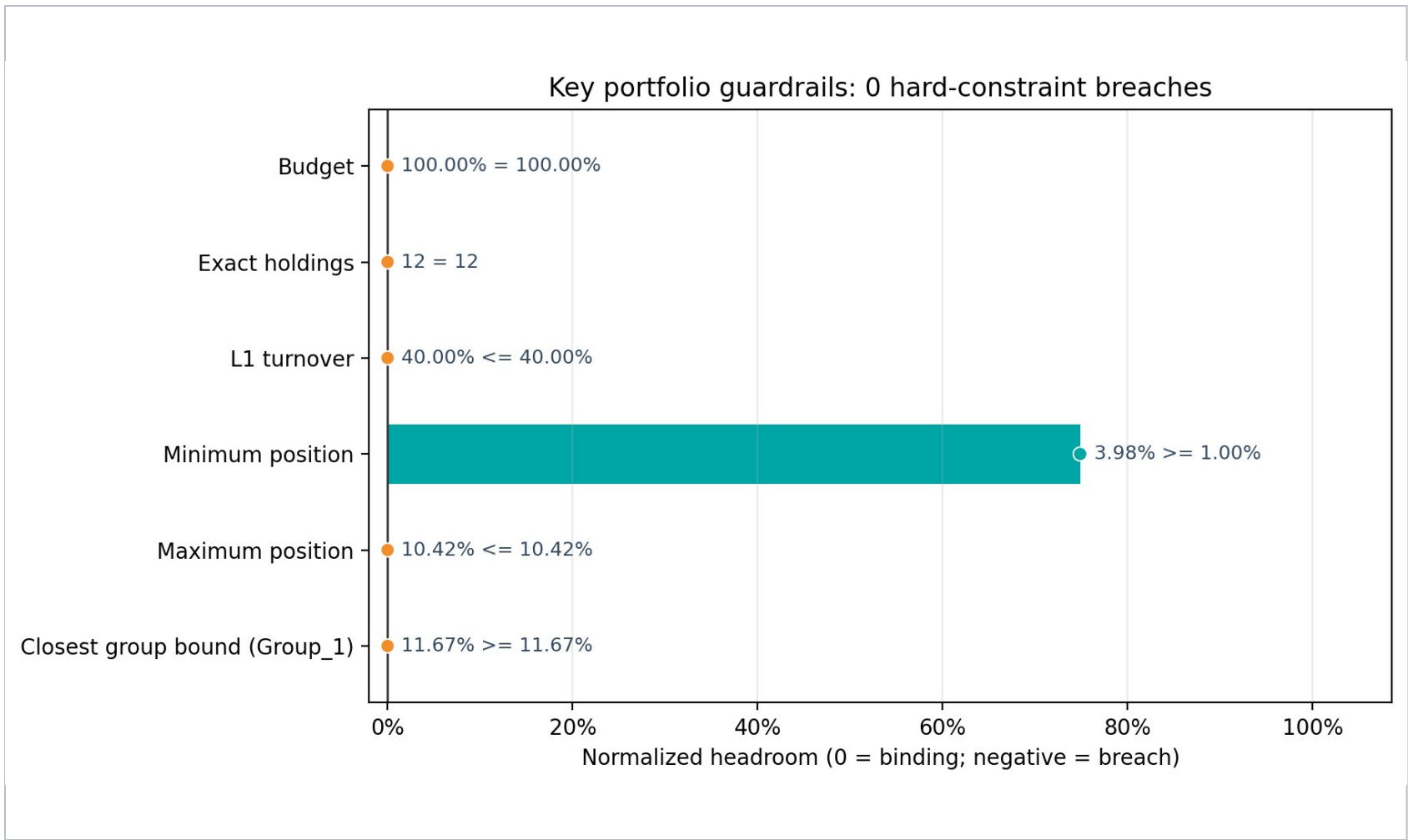
same best heuristic support after allocation

Gurobi MIQP

-0.03841471

global optimum for the stated model

All 17 guardrail families pass a global certificate



17 / 17

constraint families pass

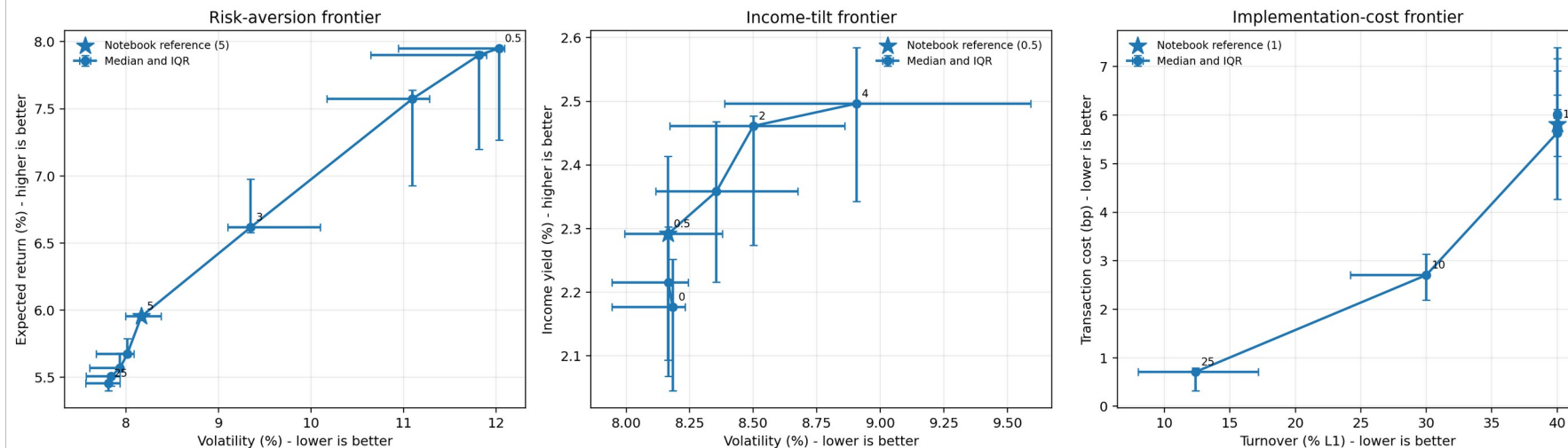
244

independent row-level checks

Gurobi objective $-0.0320099372 \cdot K = 12$ · reported gap 0.0%

Investor controls move the portfolio as expected

Canonical objective-coefficient sensitivity: actual model API, fixed K=20



Risk

12.03% → 7.81%
volatility

Income

2.18% → 2.50% yield

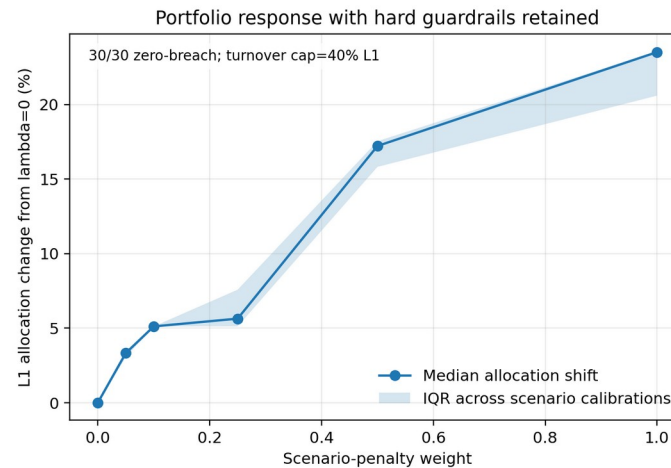
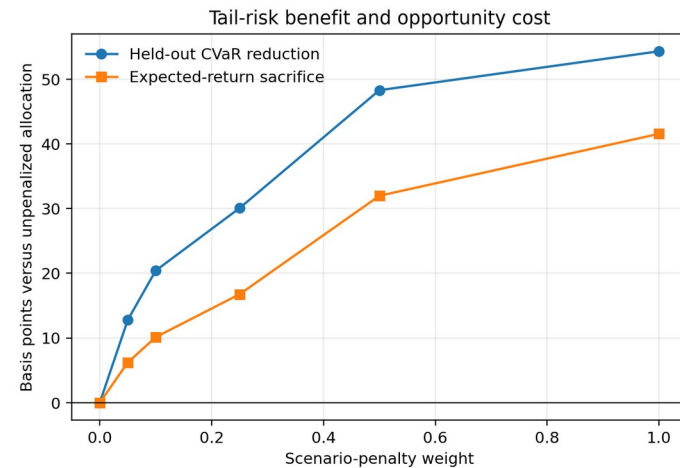
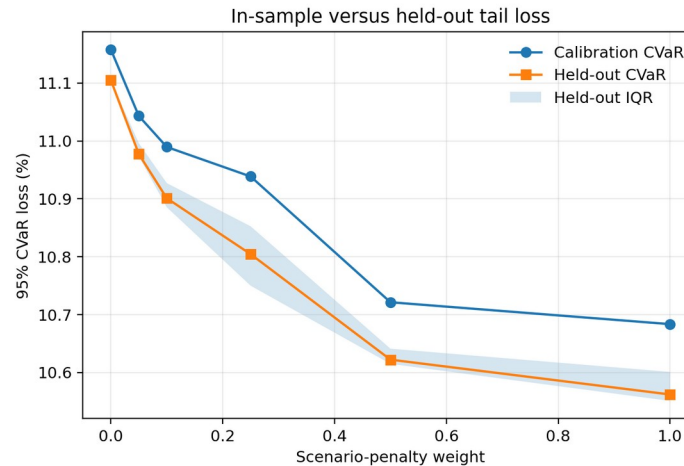
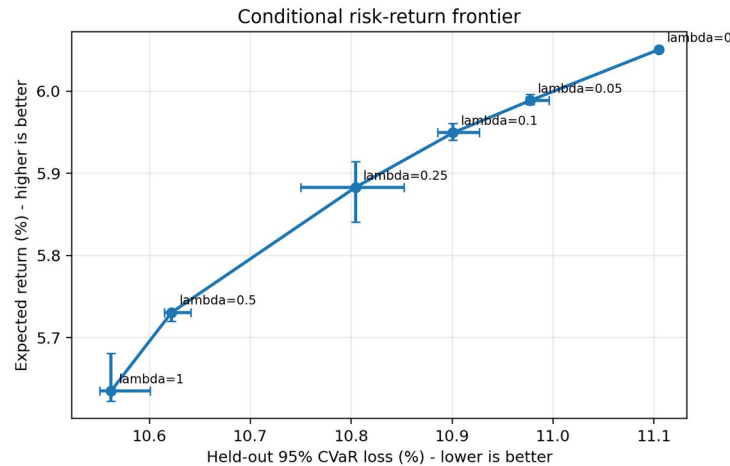
Cost

40.00% → 12.36%
turnover

Every allocation shown: zero
hard-constraint breaches

Tail risk falls 54 basis points on a clear frontier

Scenario-based CVaR preference sweep: 100 assets, fixed exact K=20



-54.31 bp

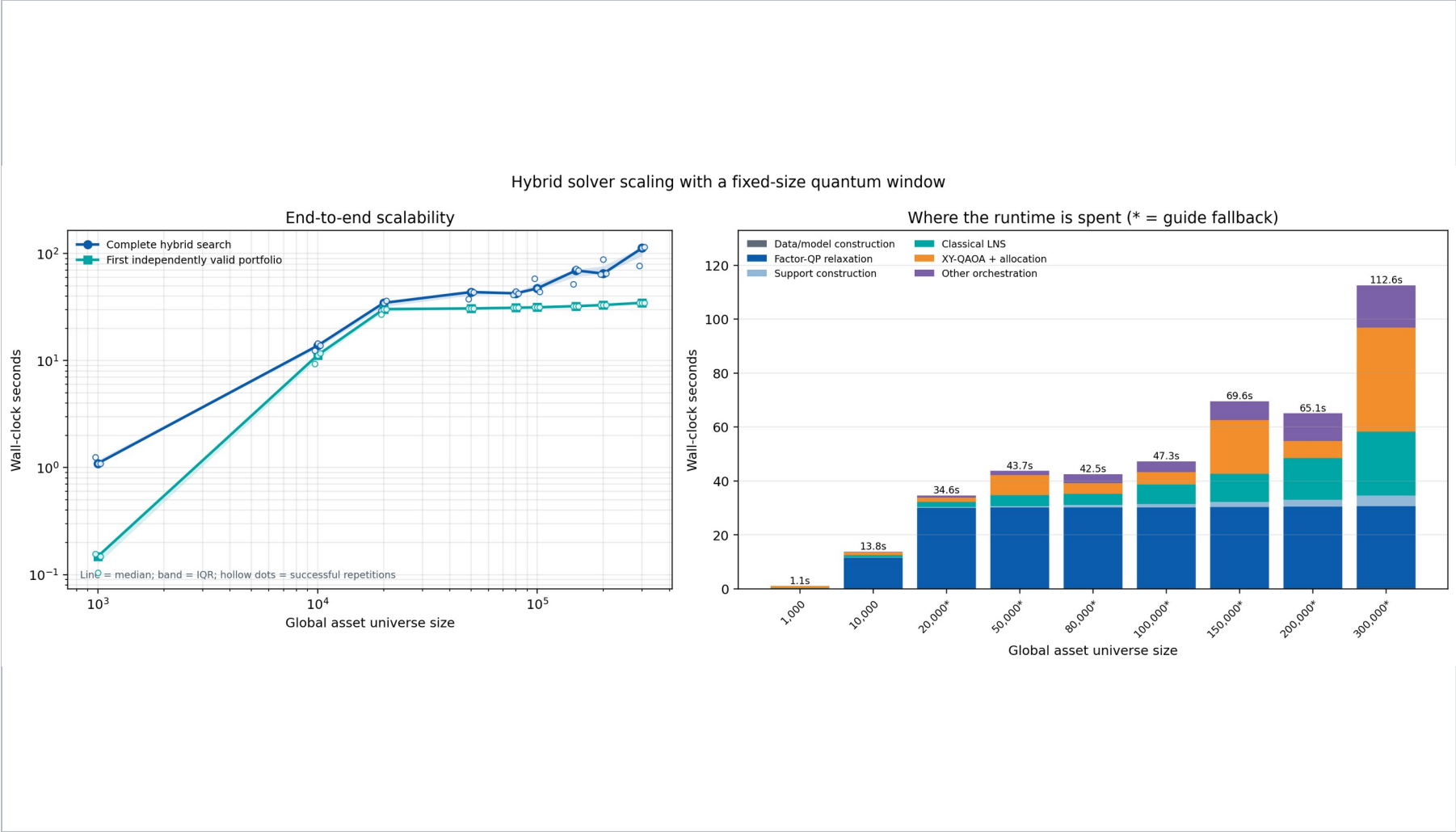
held-out 95% CVaR

-41.55 bp

expected return trade-off

30 / 30 allocations valid

Every repeated run stays valid at 300,000 assets



21 / 21

full three-window runs
valid through 20k

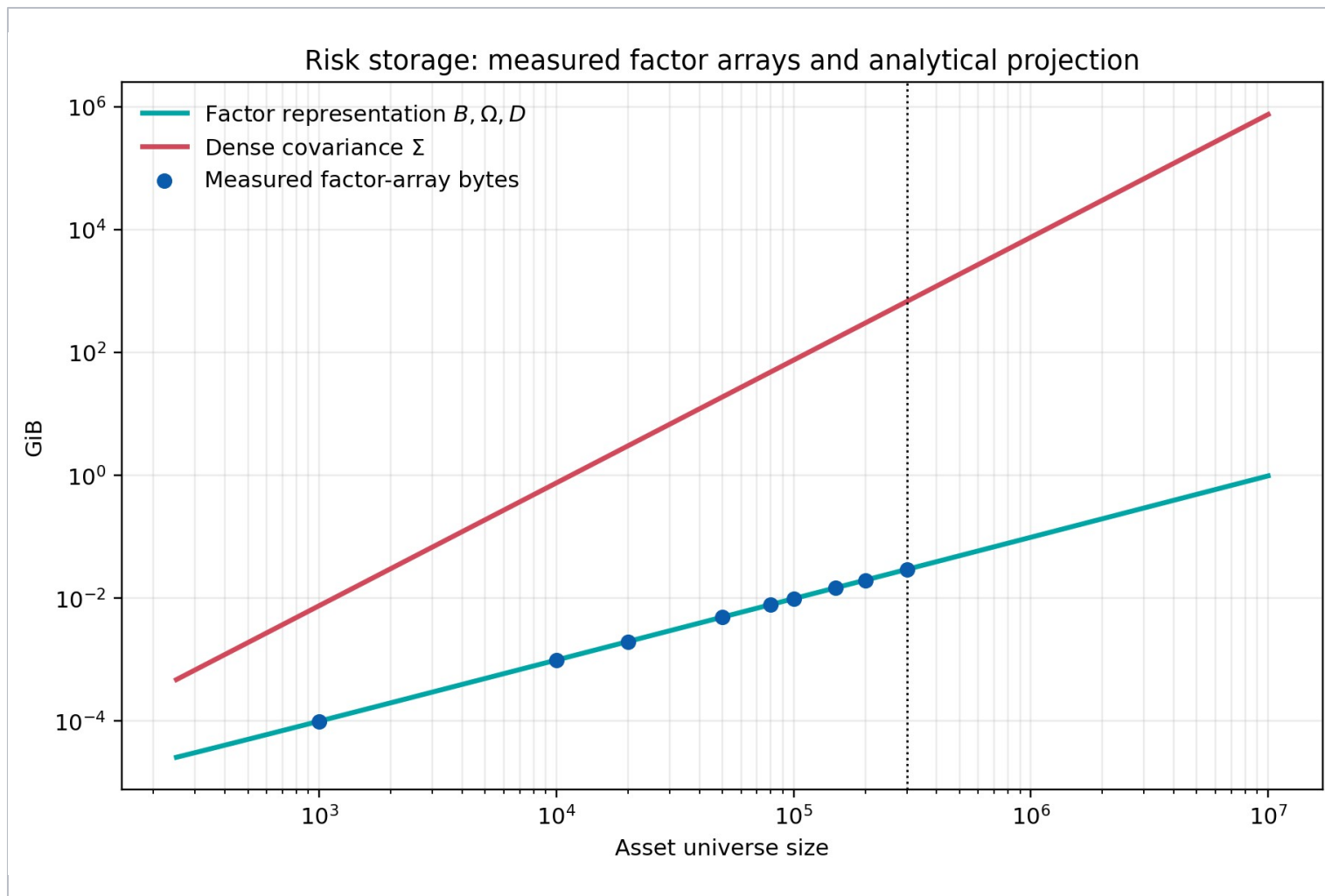
27 / 27

stretch runs valid through
300k

34.69 s

median first valid at 300k

Factor risk removes a 670 GiB memory wall



29.76 MiB

measured 12-factor arrays at
300k assets

670.55 GiB

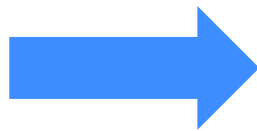
one dense covariance at 300k
assets

$\approx 23,076\times$ less risk-array storage

Quantum tests 16-asset windows—not 300k qubits

300,000

global candidate assets



16

adaptive window qubits



K = 50

validated final holdings

Exact cardinality

XY mixer exchanges $10 \leftrightarrow 01$ inside the ideal fixed-weight subspace

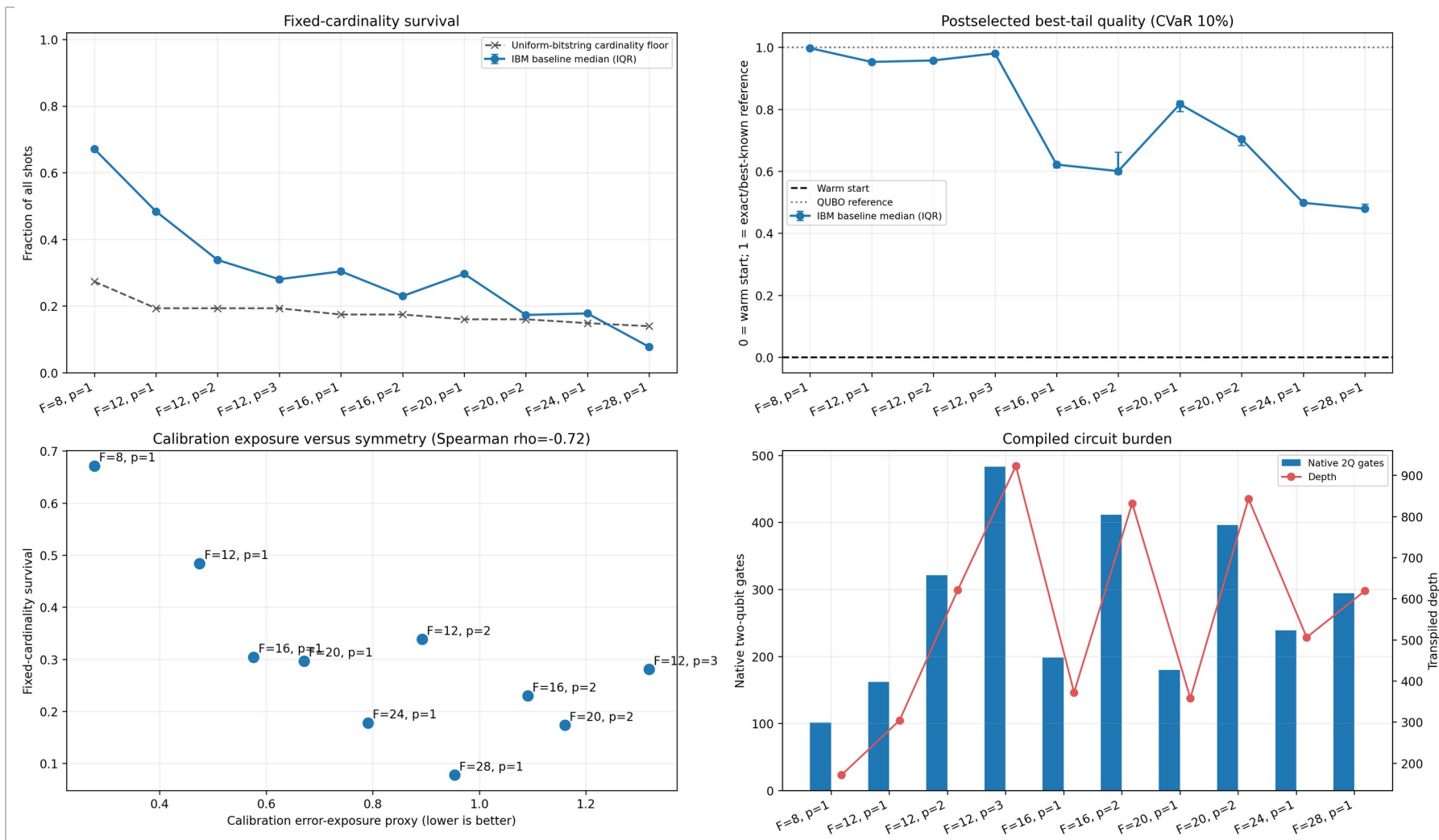
Exact allocation

Every sampled support receives continuous financial weights

Exact rejection boundary

Wrong cardinality, infeasible, duplicate, or worse candidates are rejected

IBM reaches 28 qubits; portfolio rules still hold



8–28 q

ten width-depth cases

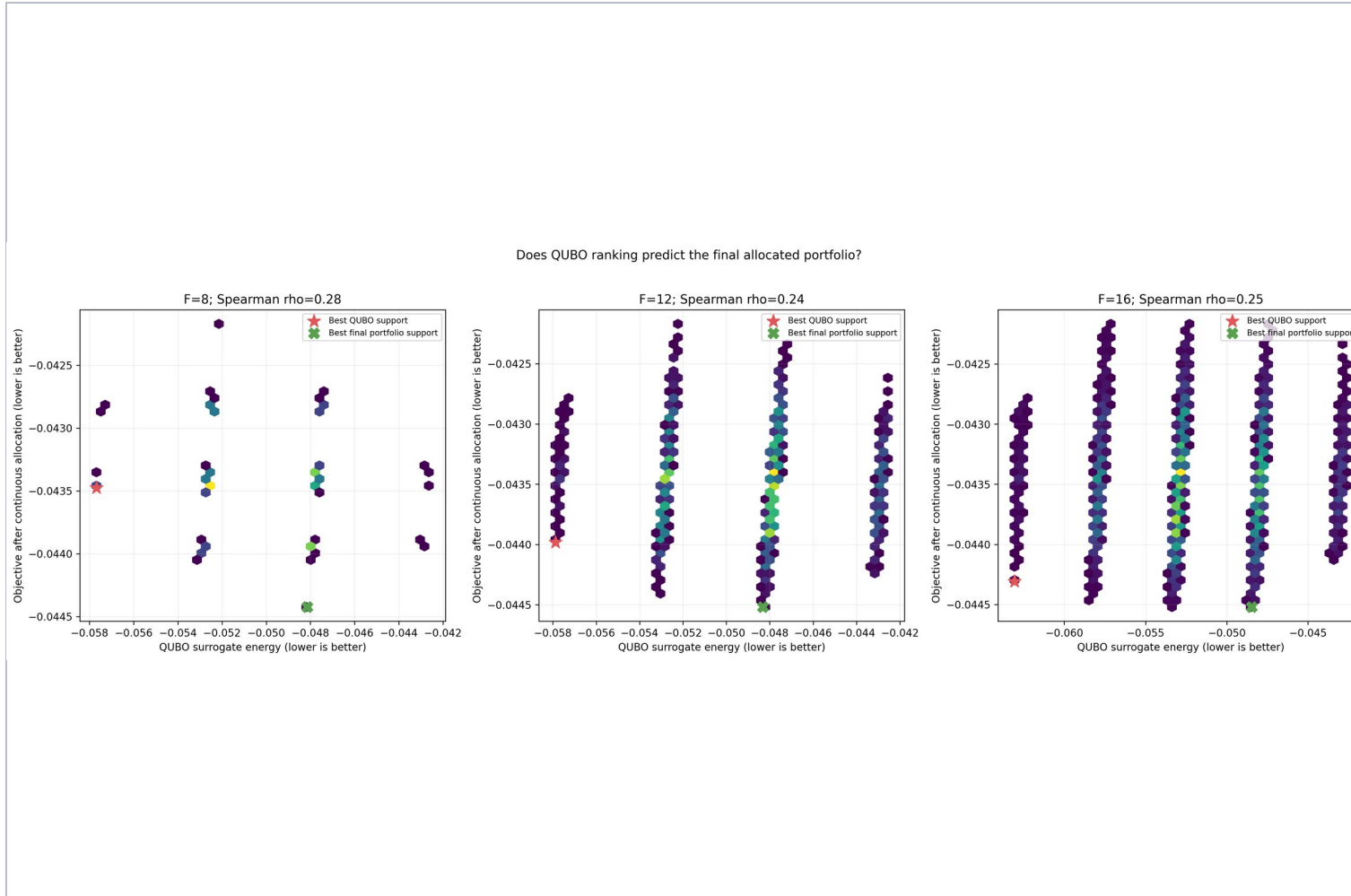
29.14%

median raw cardinality survival

30 / 30

valid after allocation + validation

The honest audit tells us exactly what to improve



0.237–0.276

Spearman QUBO ↔ allocation rank correlation

6 / 30

strict QPU wins over matched random

Roadmap

- 1 Improve the allocation-aware surrogate
- 2 Keep circuits shallow and calibrated
- 3 Match oracle-call and time budgets

Production-safe today. Quantum-ready by design.

01 Proven

Exact certificates and independent validation anchor every claim.

02 Scalable

Factor risk and local windows reach 300,000 assets without dense covariance.

03 Honest

Real IBM hardware, fair baselines, and no unsupported advantage claim.

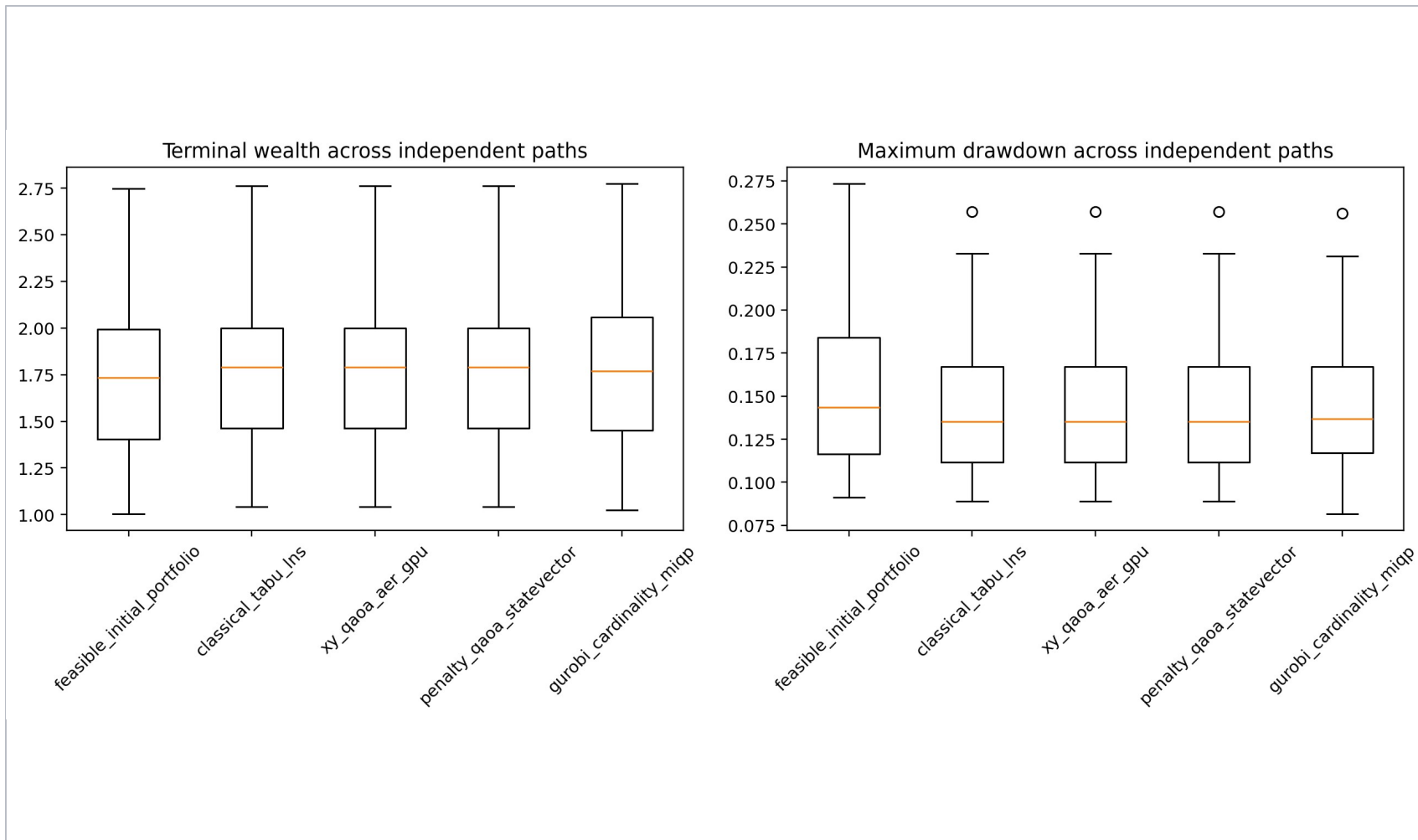
Choose the solver that can explain, validate, and reproduce every recommendation.

Appendix · Every claim has a defined experiment

Case	Scale	Runs	Quantum	Certificate	Breaches
Main sparse	100 assets	1	Aer GPU	Gurobi 0.0%	0
Full gauntlet	60 / K12	1	Aer GPU	Gurobi 0.0%	0
Scenario-rich	250 / 10k	1	16q Aer	No global	0
Full scaling	250–20k	21	16q subspace	Guide where solved	0
Stretch scaling	1k–300k	27	16q Aer GPU	Fallback above 20k	0
IBM campaign	8–28q	30 obs	ibm_kingston	Hardware only	0

Synthetic / anonymized data · seeds + environments preserved · archive SHA-256 fingerprints recorded

Appendix · In-sample optimum $\neq$ every held-out win



30 synthetic paths

LNS / XY support

Median wealth 1.7909

Volatility 8.22%

Return / vol 0.7316

Max drawdown 13.53%

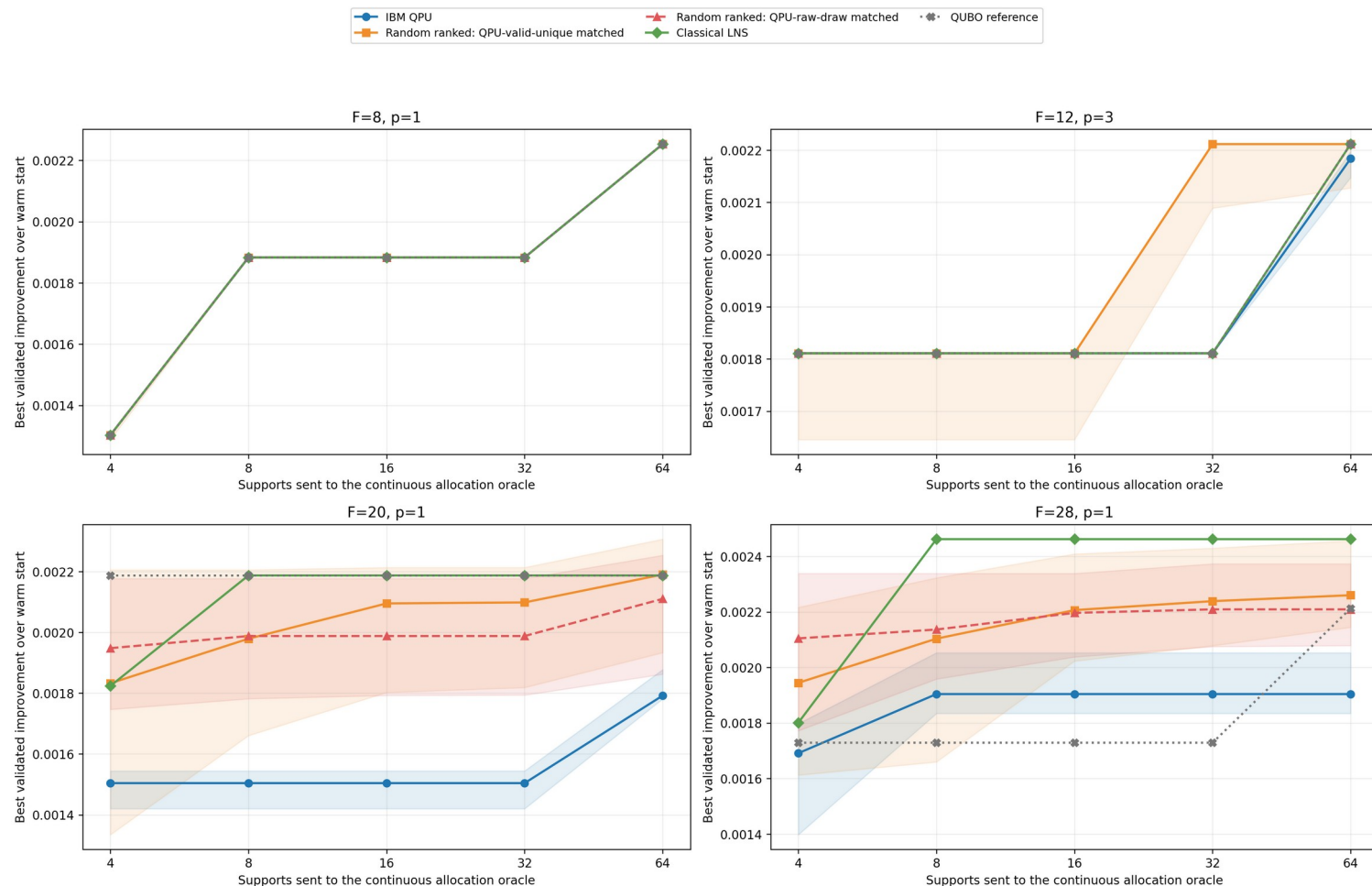
CVaR 4.23%

Gurobi global optimum

Best modeled objective,
but not every held-out
statistic

Appendix · Matched budgets make comparisons fair

Candidate quality at matched allocation budgets
Bands: QPU repetition IQR; random-pool 95% trial interval



The comparison controls

- same frozen window
- matched allocation budget
- identical oracle
- identical validator
- strict wins separated from ties

Result

Random / classical proposals usually match or beat the QPU at larger widths.

Appendix · Four labels prevent overclaiming

CERTIFIED

Enumeration or a mixed-integer incumbent and bound match

Tiny, 100-asset, 60-asset gauntlet

BOUNDED

Valid sparse portfolio compared with a solved continuous lower bound

Full-hybrid rows with solved guide

HEURISTIC

Every guardrail passes, but no global quality bound is available

10,000-scenario case and stretch scale

HARDWARE

Measured circuit and post-allocation behavior on a named QPU

IBM campaign; not an advantage claim

A fixed-support QP can be exact on one asset set without being a global sparse certificate.